

HSC Investigating Science - The Definitive Study Notes

A Comprehensive, Data-Driven Guide to the Full Syllabus

Gemini Educational Planner (Validated)

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Introduction: How to Use These Notes

This is a definitive guide for your HSC Investigating Science preparation. It is structured based on a data analysis of past HSC exams to ensure you focus on the most critical content. The information has been expanded and validated against authoritative sources like the CSIRO and NHMRC to provide the depth and accuracy you need.

Study Strategy: 1. **Master the Year 12 Modules First:** Begin with Modules 5, 8, 7, and 6. This is where the majority of HSC marks are awarded. 2. **Use the Inquiry Questions:** Each module is broken down by its inquiry questions. Use these to frame your understanding. 3. **Memorise the Glossary:** The glossary at the end contains precise, HSC-ready definitions. Use them in your answers.

Year 12 High-Priority Content

Module 5: Scientific Investigations

Inquiry question: *How are rigorous scientific investigations designed and conducted?*

This is the most important module. The concepts here are the toolkit for answering almost all Section II questions.

1. Designing Investigations

A rigorous investigation must be a **fair test** that directly addresses the hypothesis.

- **Hypothesis:** A testable statement that proposes a relationship between variables. It must be specific and able to be either supported or refuted by the evidence you collect.
- **Variables:**
 - **Independent Variable (IV):** The single factor that you deliberately change or manipulate to see what effect it has.
 - **Dependent Variable (DV):** The factor that you measure or observe to see if it has been affected by the change in the IV.
 - **Controlled Variables:** All other factors that could potentially influence the DV, which must be kept constant to ensure a fair test.
- **Control Group:** A group in an experiment that does not receive the treatment (the IV is not applied). It acts as a baseline, showing what would happen without any intervention, allowing you to be more certain that any changes in the experimental group are due to the IV.

2. Ensuring Data Quality: Validity, Reliability, Accuracy

These three terms are distinct and crucial.

- **Validity:** An experiment is **valid** if its method is appropriate for testing the hypothesis. It is the most important aspect of an investigation.
 - **Threats to Validity:** Uncontrolled variables, biased sampling, or a method that doesn't actually measure the dependent variable.
 - **Ensuring Validity:** Meticulously control all variables. Use a control group. Ensure your measurement technique is directly assessing the effect of the IV on the DV.
 - **Example:** An experiment testing if fertiliser makes a plant grow taller is *not valid* if the fertilised plant also gets more sunlight. Sunlight becomes an uncontrolled variable.
- **Reliability:** The results of an experiment are **reliable** if they are consistent and repeatable.
 - **Threats to Reliability:** Random errors, insufficient data, or a small sample size.
 - **Ensuring Reliability:** Repeat the experiment multiple times (at least 3) and check for consistency. If the results are similar each time, they are reliable. Calculating an average from repeated trials is a key way to increase reliability.
 - **Example:** Measuring the time it takes for a ball to drop once is not very reliable. Measuring it five times and finding the times are all very close (e.g., 1.2s, 1.21s, 1.19s, 1.2s, 1.22s) makes the result reliable.
- **Accuracy:** A measurement is **accurate** if it is close to the true, accepted value.
 - **Threats to Accuracy:** Systematic errors from faulty or uncalibrated equipment.
 - **Ensuring Accuracy:** Use high-quality, precise measuring instruments. Calibrate them before use (e.g., zero a scale). Minimise human error by using correct techniques (e.g., reading a measuring cylinder at eye level to avoid parallax error).
 - **Example:** A thermometer that consistently reads 2°C too high is reliable (it gives the same wrong reading every time) but not accurate.

3. Data Analysis and Communication

- **Data Types:**
 - **Qualitative:** Descriptive (e.g., colour, smell, texture).
 - **Quantitative:** Numerical (e.g., 10.5 cm, 25°C, 50 grams).
- **Presenting Data:** Use tables for raw data and graphs to show trends and relationships. Always include a title, labelled axes with units, and a key/legend if needed.

- **Scientific Reports:** Follow a standard structure: Title, Abstract, Introduction (with hypothesis), Method, Results, Discussion, Conclusion. The language should be objective, formal, and written in the third person.
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Module 8: Science and Society

Inquiry question: *What is the role of science in society and how is it influenced?*

1. Ethics in Research

Scientific research must be conducted ethically to protect participants and ensure public trust. In Australia, the **NHMRC (National Health and Medical Research Council)** sets the standards.

- **Core Ethical Principles (NHMRC):**
 - **Respect:** Respecting the intrinsic value of individuals. This leads to the requirement for **informed consent**, where participants know the risks and benefits before agreeing to take part.
 - **Research Merit and Integrity:** The research must be justifiable, well-designed, and conducted honestly.
 - **Justice:** The benefits and burdens of research must be distributed fairly. No group should be exploited or unfairly excluded.
 - **Beneficence:** The likely benefits of the research must justify any potential risks of harm or discomfort to participants.
- **Key Case Studies:**
 - **Nuremberg Code:** Developed after WWII in response to unethical Nazi experimentation. It established the principle of voluntary consent as essential.
 - **Tuskegee Syphilis Study:** A highly unethical study where African American men with syphilis were not told they had the disease and were not given treatment, even after penicillin became available. This led to major reforms in research ethics.

2. Influences on Science

- **Economic:** Funding is the lifeblood of research.
 - **Government Funding (e.g., ARC):** Governments direct funding to areas of national priority (e.g., health, environment, technology). This can shape the direction of a country's entire research landscape.

- **Corporate Funding:** Private companies fund research to develop new products. This can lead to a **conflict of interest**, where the need to produce a profitable result could bias how research is conducted or reported.
 - **Social and Political:**
 - **Public Opinion & Social Need:** Major societal issues (e.g., pandemics, climate change, disease outbreaks) create immense pressure for scientific solutions, leading to massive funding and collaboration.
 - **Political Agendas:** Government policies and international agreements can either promote or restrict certain types of research (e.g., funding for renewable energy vs. fossil fuels, or regulations on genetic research).
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Module 7: Fact or Fallacy?

Inquiry question: *How can a claim be tested?*

1. Evaluating Evidence and Claims

- **Correlation vs. Causation:** This is a critical distinction.
 - **Correlation:** A relationship where two variables change together.
 - **Causation:** A relationship where a change in one variable *causes* a change in another.
 - **The Trap:** Just because two things are correlated does not mean one causes the other. There could be a third, **confounding variable** responsible.
 - **Example:** There is a correlation between coffee drinking and lung cancer. The confounding variable is smoking; smokers are more likely to drink coffee and smoking causes lung cancer. When you control for smoking, the correlation disappears.
- **Evidence-Based Claims:** A strong scientific claim is backed by reliable, valid, and verifiable evidence from well-designed investigations, preferably published in peer-reviewed journals.
- **Pseudoscience:** Claims that appear scientific but are not based on the scientific method.
 - **Red Flags:** Vague or untestable claims; reliance on anecdotes (“it worked for me”) instead of data; confirmation bias (ignoring contradictory evidence); lack of peer review.

2. The Role of the Scientific Community

- **Peer Review:** The process where a scientific paper is sent to other experts in the field for anonymous evaluation before it is published. It is a crucial quality control mechanism to check for validity, significance, and originality. It does not guarantee the work is correct, but it ensures it has met a high standard of scientific rigour.
 - **Cognitive Biases:**
 - **Confirmation Bias:** The tendency to favour information that confirms one's pre-existing beliefs.
 - **Halo Effect:** The tendency for a positive impression of a person in one area to positively influence one's opinion of them in other areas (e.g., believing a famous physicist's opinion on nutrition).
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Module 6: Technologies

Inquiry question: *How does the development of technologies affect scientific investigation?*

1. The Interdependent Relationship

Science and technology have a feedback loop.

- **Science -> Technology:** Scientific discoveries provide the foundational knowledge for technological breakthroughs.
 - **Example:** The discovery of the properties of semiconductors (science) led to the invention of the transistor (technology), which is the basis of all modern electronics.
- **Technology -> Science:** New technologies provide more powerful tools for scientists to make new discoveries.
 - **Example:** The invention of the telescope (technology) allowed Galileo to observe the moons of Jupiter, providing evidence that disproved the geocentric model of the universe (science).

2. Impact of Technology

Be prepared to discuss a specific technology and its impact. * **Example: GPS (Global Positioning System)** * **Scientific Origin:** Based on Einstein's Theory of General Relativity. The clocks on GPS satellites must be adjusted to account for relativistic effects to maintain accuracy. * **Impact:** Revolutionised navigation, logistics, agriculture, and emergency services.

Year 11 Foundational Content

Module 1 & 2: Cause and Effect

- **Observation:** Gathering data directly using senses or instruments.
- **Inference:** A logical conclusion based on observations.
- **Generalisation:** A broad statement or conclusion drawn from specific cases.
- **Types of Reasoning:**
 - **Deductive:** Starts with a general principle to make a specific conclusion (e.g., All birds have feathers. A robin is a bird. Therefore, a robin has feathers).
 - **Inductive:** Uses specific observations to form a general conclusion (e.g., I have observed 100 swans, and they are all white. Therefore, all swans are white). Inductive reasoning is powerful but can be proven wrong by new evidence (e.g., finding a black swan).

Module 3: Scientific Models

- **Purpose:** To simplify and represent complex systems, make predictions, and communicate ideas.
- **Types:**
 - **Physical:** A tangible representation (e.g., a DNA double helix model).
 - **Mathematical:** Uses equations (e.g., $F=ma$).
 - **Conceptual:** A diagram or flowchart (e.g., a food web).
- **Limitations:** A model is a simplification and may not account for all variables or complexities of the real world.

Module 4: Theories and Laws

- **The Scientific Method:** A cyclical process of observation -> hypothesis -> experimentation -> analysis -> conclusion -> communication. The results often lead to new questions and further investigation.
- **Key Distinction:**
 - A **Law** *describes* what happens (e.g., Newton's Law of Universal Gravitation describes the force between two objects).
 - A **Theory** *explains why* it happens (e.g., Einstein's Theory of General Relativity explains that gravity is the result of mass warping spacetime).

Complete Glossary of Key Terms

Term	Definition
Accuracy	How close a measured value is to the true or accepted value.
Bias	A systematic error or prejudice that can influence the results of an investigation, leading to a conclusion that is not objective.
Causation	The relationship where a change in one variable is directly responsible for a change in another variable.
Conflict of Interest	A situation in which a person or organisation has competing interests (e.g., financial) that could potentially compromise their impartiality or objectivity in conducting or reporting research.
Control Group	A group in an experiment that is not exposed to the independent variable, used as a baseline to compare the results of the experimental group.
Controlled Variable	A variable that is kept constant throughout an experiment to ensure a fair test.
Correlation	A mutual relationship or connection between two or more things, where one variable changes in relation to another.

Term	Definition
Dependent Variable	The variable that is measured or observed in response to the change in the independent variable.
Double-Blind Trial	An experiment in which neither the participants nor the researchers know who is receiving a particular treatment, used to prevent bias.
Ethics	Moral principles that govern a person's or group's behaviour, concerning what is right and wrong in the conduct of scientific research.
Hypothesis	A testable prediction about the relationship between variables, which can be supported or refuted by evidence from an investigation.
Independent Variable	The variable that is deliberately changed or manipulated by the investigator.
Inference	A logical conclusion based on observations and prior knowledge.
Law (Scientific)	A statement that describes an observed phenomenon or a pattern in nature, often expressed as a mathematical relationship, without explaining why it occurs.
Model (Scientific)	A simplified representation of a system, object, or phenomenon that helps scientists understand and predict its behaviour.
Observation	The process of gathering information about events or processes in a careful, orderly way, using the senses or scientific instruments.
Outlier	A data point that is significantly different from other observations in a data set.
Peer Review	The evaluation of scientific work by other experts in the same field to ensure its quality and validity before it is published.
Placebo	A substance or treatment with no active therapeutic effect, used as a control in clinical trials to test the effectiveness of a new treatment.
Precision	How close multiple measurements of the same quantity are to each other.

Term	Definition
Pseudoscience	A claim, belief, or practice presented as scientific, but which does not adhere to the scientific method and lacks supporting evidence.
Qualitative Data	Descriptive data that is observed, not measured (e.g., colour, texture, smell).
Quantitative Data	Numerical data that is measured or counted (e.g., length, mass, time, temperature).
Random Error	Unpredictable variations in measurements that occur by chance and affect the precision of the results.
Reliability	The extent to which the results of an experiment are consistent and repeatable.
Systematic Error	An error that is consistent and repeatable, often caused by faulty equipment or a flawed experimental method, which affects the accuracy of the results.
Technology	The application of scientific knowledge for practical purposes, especially in industry, involving the development of devices, methods, and systems.
Theory (Scientific)	A well-substantiated explanation of some aspect of the natural world, based on a body of facts that have been repeatedly confirmed through observation and experiment.
Validity	The extent to which an experiment tests the intended hypothesis (i.e., it is a ‘fair test’).